Microeconomics I

Class 7 — The Slutsky equation
and the money value of a price change

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Slide 1 Title

Class 7 closes the circle of classes 4 to 6. The two demands, x from the budget and h from the curve, are tied together by the Slutsky equation, which splits any price effect into a move along the indifference curve and a wealth effect; then the whole map of x, v, h and e; and the payoff, what a price change is worth to the consumer in euros, measured at the old prices or at the new. MWG 3.G and 3.I.

Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letters of your answers in the block on the sheet, write your name on it, and hand in only the strip.

Plan for today

10:00	Short quiz
10:10	Warm-up; a mug, three prices
10:20	The Slutsky equation: substitution and wealth effects
10:35	Giffen goods, revisited
10:40	In pairs: gross and net cross effects
10:50	Break
11:00	The Slutsky matrix; the map of x, v, h, e
11:10	Compensating and equivalent variation (CV, EV)
11:29	In pairs: a cheaper good
11:38	Deadweight loss; summary

Slide 2 Plan for today

Before the break: last week's price rise taken apart into a substitution effect and a wealth effect; the Slutsky equation, which does the same split for a small change; what it says about market against compensated demand and about Giffen goods; and an exercise on class 6's gross-versus-net table. After the break: the Slutsky matrix from observable demand, the map of x, v, h, e, and the money value of a price change, compensating and equivalent variation, with a tax as the last example.

MWG 3.G and 3.I; Muñoz-Garcia chapter 2 is the gentler read.

Practicalities

- Problem set 2: due at the start of class, Wed 21 Oct — on paper, one copy per group
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Slide 3 Practicalities

Problem set 2 is due at the start of class on Wednesday 21 October: the whole booklet, one per group, pages in order and stapled, all names in the box on the cover and nowhere else; extra sheets go at the end, each marked with the group and the part. Before you hand it in, photograph the code on the cover: once the booklet is marked, it opens your marked problem set. Today's material, the Slutsky equation and the money measures, is not on it.

Office hours are on Wednesdays, 14:00–15:00, Room 445, up to 20 minutes per student; booking by e-mail gives priority. Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Warm-up: always true, or not?

Last class. True for every consumer? One minute with your neighbour.

p_1 rises $\Rightarrow$ $h_1(p,u)$ does not rise **true: compensated law of demand**	p_1 rises $\Rightarrow$ $x_1(p,w)$ does not rise **false: Giffen (class 5's rice)**
$e(p,u)$ is homogeneous of degree 0 in p **false: degree 1**	$e\big(p, v(p,w)\big) = w$ **true: duality**

h_1 never rises with p_1; x_1 can. Today: the difference is a wealth effect.

Class 6, slides 10, 13, 14; class 5, slide 22. MWG Prop. 3.E.2, 3.E.4; (3.E.1).

A mug, three groups: whose prices should be equal?

Simon Fraser University: 77 students, a coffee mug, three groups at random

Sellers are given a mug: the lowest price at which they sell it

Buyers: the highest price they pay for one

Choosers: a mug or cash: the cash that is worth as much as the mug

Which two medians should standard theory make equal?

buyers and choosers · choosers and sellers · buyers and sellers

Kahneman, Knetsch & Thaler (1990), Journal of Political Economy 98(6), experiment 6.

Slide 4 Warm-up: always true, or not?

Four one-line claims from last week, each about one of class 6's objects, and the question is whether each holds for every consumer. The two in the top row look alike and differ only in what is held fixed: utility for h, wealth for x. Compensated demand can never rise in its own price, because the cheapest bundle at the new prices costs at most what the old one costs there (class 6, slide 14); market demand can, as class 5's rice consumer showed, because a price change also moves her purchasing power.

The bottom row is about e: doubling all prices doubles the cost of every bundle and so of the cheapest one, degree 1, not the degree 0 of h in p (and of v in p and w together); and the least cost of the utility that w buys is exactly w, the identity behind the name compensated. The gap between the two lines of the top row is what the first hour measures: the wealth effect of a price change. The trap is to treat x and h as one function; they agree only at the point where $u = v(p,w)$.

Slide 5 A mug, three groups: whose prices should be equal?

A laboratory experiment from 1990 with 77 students at Simon Fraser University, assigned at random to three groups: sellers, who were given a mug and named the lowest price at which they would part with it; buyers, who named the highest price they would pay for one; and choosers, who were offered a mug or cash and named the cash that was worth as much as the mug. Each group gives a median price, so one object is priced in three ways.

The question is which two of the three should coincide for a consumer as in our theory. Choosers and sellers both end up with either a mug or money; buyers start without a mug. What the data said comes on the next slide; the theory that answers the question, the two money measures of a change, comes after the break.

Source: Kahneman, Knetsch and Thaler (1990), Journal of Political Economy 98(6), experiment 6.

Choosers and sellers face the same choice: 3.12 vs 7.12

Buyers	Choosers	Sellers
$2.87	**$3.12**	**$7.12**
pay at most	cash worth a mug	accept at least

medians, 77 students

Choosers and sellers both choose between a mug and cash, yet sellers ask more than twice as much. What theory allows: after the break.

Kahneman, Knetsch & Thaler (1990), "Experimental tests of the endowment effect and the Coase theorem", JPE 98(6): 1325–1348, experiment 6, p. 1339.

Slide 6 Choosers and sellers face the same choice: 3.12 vs 7.12

The three medians: buyers would pay at most $2.87, choosers wanted $3.12 in cash to forgo a mug, sellers asked at least $7.12. Buyers and choosers nearly agree. Choosers and sellers face the same final choice, a mug or cash, and the only difference between them is that the sellers were handed the mug; yet their median is more than twice the choosers'.

The slide only states the puzzle. Whether consumer theory can produce either gap, and how large a gap it allows, needs the two money measures of a change defined after the break; slide 22 returns to these numbers. The authors' own reading is that the sellers' and choosers' positions were identical, so no difference in wealth can explain the gap.

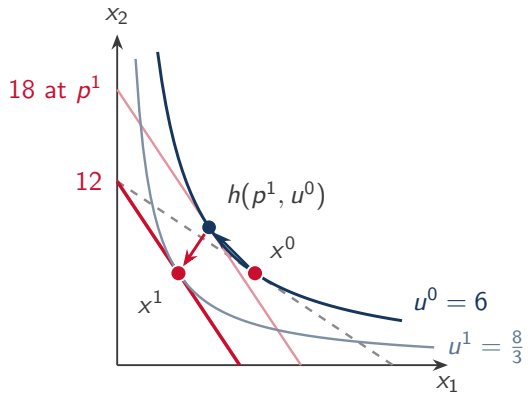
The Slutsky equation

MWG 3.G

Slide 7 The Slutsky equation

Part 1: the Slutsky equation, which splits a price effect into a substitution effect and a wealth effect. MWG 3.G.

Good 1 dearer: x_1 falls $3 \to 2$ along u^0, then $2 \to \frac{4}{3}$ from lost wealth



x_1: $3 \to 2 \to \frac{4}{3}$

substitution -1, wealth $-\frac{2}{3}$

x_2: $2 \to 3 \to 2$

substitution $+1$, wealth -1

18 euros at p^1 keep her on u^0

Substitution: along u^0, at the new prices.

Wealth: the parallel shift from 18 euros down to her 12.

MWG 3.G, pp. 71–72; class 6, slide 14. $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.

Slide 8 Good 1 dearer: x_1 falls $3 \to 2$ along u^0, then $2 \to \frac{4}{3}$ from lost wealth

Class 6's price rise, good 1 from 2 to 4.5 with 12 euros, moved her from $(3, 2)$ to $(\frac{4}{3}, 2)$. The slide splits that move at an intermediate bundle: the cheapest bundle on her old curve at the new prices, $h(p^1, u^0) = (2, 3)$, which 18 euros would buy (class 6, slide 22). The first leg, along u^0, is the substitution effect: she reacts to the new relative price while held exactly as well off, so it always runs away from the dearer good. The second leg, from the 18-euro line down to her actual 12-euro line, is a parallel shift at the new prices, the wealth effect, which depends on the good; here both goods are normal, so both fall.

For good 2 the two legs cancel exactly because $x_1 x_2$ spends half of w on each good whatever the prices: a feature of this consumer, not a rule. MWG compensates with utility held fixed, the Hicksian decomposition; the next slide writes the same split for a tiny price change. The trap is to draw the substitution step at the old prices: it is the new relative price along the old curve.

Slutsky equation: compensated slope = market slope + wealth term

$$\frac{\partial h_\ell(p,u)}{\partial p_k} = \frac{\partial x_\ell(p,w)}{\partial p_k} + \frac{\partial x_\ell(p,w)}{\partial w} x_k(p,w) \quad \text{at } u = v(p,w)$$

why: differentiate $h_\ell(p,u) = x_\ell\big(p, e(p,u)\big)$ in p_k; $\partial e/\partial p_k = h_k = x_k$ (Shephard)

$$\frac{\partial x_\ell}{\partial p_\ell} = \underbrace{\frac{\partial h_\ell}{\partial p_\ell}}_{\leq 0:\ \text{substitution}} \underbrace{- x_\ell \frac{\partial x_\ell}{\partial w}}_{\text{wealth}}$$

$x_1 x_2$ at $p = (2, 3)$, $w = 12$: $-\frac{3}{2} = -\frac{3}{4} - 3 \cdot \frac{1}{4}$

A normal good falls more in market demand than in compensated demand.

MWG Prop. 3.G.3, p. 71, (3.G.3)–(3.G.4): u continuous, $\succsim$ locally non-satiated and strictly convex.

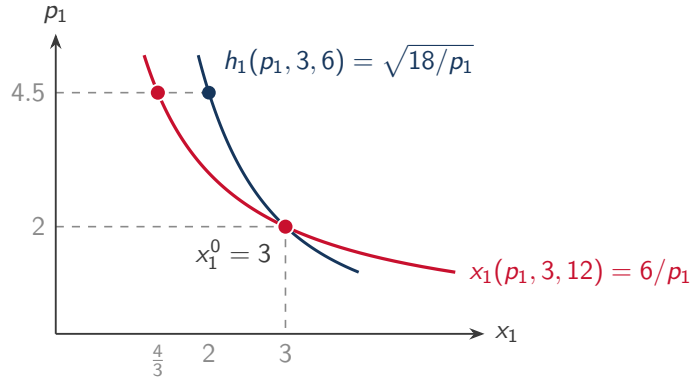
Slide 9 Slutsky equation: compensated slope = market slope + wealth term

Read the first line as a sentence: the slope of compensated demand in a price equals the slope of market demand plus a wealth correction, the wealth slope times the quantity of the good whose price moved. The proof needs only class 6: h is market demand at wealth $e(p, u)$, and when p_k rises by a cent the cost of staying on the curve rises by h_k cents (Shephard's lemma), which is x_k at $u = v(p, w)$. So x_k is how much poorer a one-cent rise in p_k makes her, to first order.

The own-price form is slide 8 in derivatives: market effect = substitution effect, never positive, minus quantity times wealth slope. For $x_1 x_2$ at $(2, 3)$ and 12 the substitution part comes out as $-\frac{3}{4}$, class 6's compensated slope from its slide 20, the first check that the equation ties the two demands correctly; slide 8's -1 and $-\frac{2}{3}$ were the large-change version of the same two pieces. The slip is to put the wrong quantity in the wealth term: it is the good whose price moves, x_k, not the good whose demand we watch.

Market demand falls faster than compensated: good 1 is normal

$u = x_1 x_2$, $p_2 = 3$, $w = 12$, $u^0 = 6$: both curves pass through $x_1 = 3$ at $p_1 = 2$



At $p_1 = 4.5$: compensated demand 2, market demand $\frac{4}{3}$; the gap is the wealth effect.

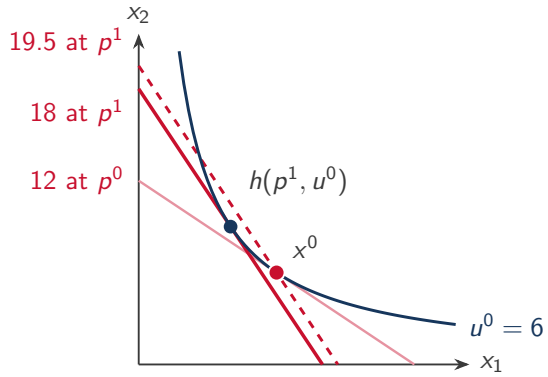
MWG Figure 3.G.1(a), p. 71; text p. 72.

Slide 10 Market demand falls faster than compensated: good 1 is normal

MWG draws demand with the price on the vertical axis, so a curve that reacts more to price looks flatter. The two curves share the point $x_1 = 3$ at $p_1 = 2$ because at that price twelve euros buy exactly $u = 6$, class 6's identity $e(p, v(p, w)) = w$. Away from that price they separate: above 2 a dearer good 1 makes her poorer and, good 1 being normal, she buys even less than compensated demand would, so the market curve lies to the left; below 2 a cheaper good makes her richer and the market curve lies to the right.

Why it matters: the whole gap between the curves is the wealth term of slide 9, and its sign is the sign of $\partial x_1/\partial w$. For an inferior good the two curves change places (MWG Figure 3.G.1(b)), and a Giffen good is the extreme case where the market curve bends the other way, slide 12. The trap is to say that the compensated curve is always the steeper one without saying for which goods.

Two ways to compensate: keep u^0 (6 euros) or x^0 affordable (7.5)



Hicks: $e(p^1, u^0) - w = 18 - 12 = 6$
back on her old curve

Slutsky: $p^1 \cdot x^0 - w = 19.5 - 12 = 7.5$
her old bundle affordable again

Hicks $\leq$ Slutsky, equal to first order: so the compensated slopes of slide 9 carry Slutsky's name.

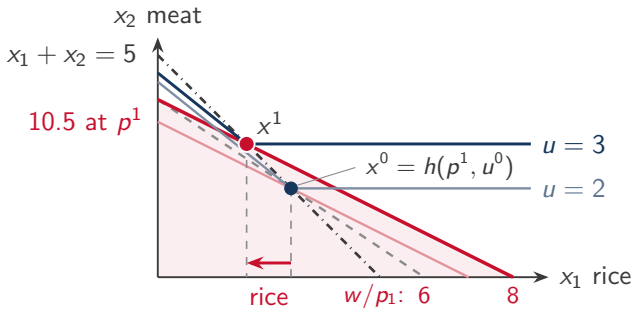
MWG 3.G, pp. 72–73, Figure 3.G.2; class 6, slide 22.

Slide 11 Two ways to compensate: keep u^0 (6 euros) or x^0 affordable (7.5)

Two ways to make up for the price rise. Hicks gives her enough to get back to her old indifference curve, $18 - 12 = 6$ euros; Slutsky gives her enough to afford her old bundle again, $19.5 - 12 = 7.5$, class 6's fixed basket. Hicks can never cost more, because the old bundle is one way to reach u^0, so the cheapest way costs at most as much; with 19.5 euros she would not keep $(3, 2)$ but substitute and end above u^0.

For a tiny price change the two agree to first order, since both equal $x^0 \cdot \Delta p$: the price gradient of e at the start is $h(p^0, u^0) = x^0$ by Shephard's lemma. That is why the compensated slopes of slide 9, built with Hicksian compensation, carry Slutsky's name. The trap is to think the two compensations coincide for any size of change; they coincide only in the limit.

Giffen needs inferior: class 5's rice, $3 \to 3 \to 2$



rice p_1: $2 \to 1.5$

x_1: $3 \to 3 \to 2$

substitution 0 (the kink on the food line $x_1 + x_2 = 5$), wealth -1

at $p_1 = 2$:

$$\frac{\partial x_1}{\partial p_1} = 0 - 3 \cdot (-1) = +3$$

$\partial h_1/\partial p_1 = 0$, $x_1 = 3$, $\partial x_1/\partial w = -1$

Substitution is never positive, so $\partial x_\ell/\partial p_\ell > 0$ needs $\partial x_\ell/\partial w < 0$: Giffen needs inferior.

Class 5, slides 19 and 22: $u = \min\{x_2,\ x_2 - 4(5 - x_1 - x_2)\}$, $w = 12$, $p_2 = 3$. MWG Prop. 3.G.3, p. 71.

Slide 12 Giffen needs inferior: class 5's rice, $3 \to 3 \to 2$

Class 5's promise. The rice consumer bought less rice when it got cheaper, and the decomposition shows why that is possible. Her indifference curve has a kink at her old bundle, on the dash-dotted food line $x_1 + x_2 = 5$ where every one of her curves bends (class 5), and at the new price the compensating cost line still touches the curve only at the kink: held on her old curve she changes nothing, so the substitution effect is zero. The whole fall in rice is the wealth effect: cheaper rice leaves her 1.5 euros to spare, and richer, she buys less rice, which is inferior for her (class 5, slide 19).

In derivatives, the own-price form of slide 9 gives the general lesson: the substitution term is never positive, so a positive market slope needs a negative wealth term, an inferior good whose wealth effect is strong enough to beat substitution. The converse fails: most inferior goods are not Giffen, because their substitution effect wins. Here $\partial h_1/\partial p_1 = 0$ is the weakest substitution effect possible, and that is what the kink buys.

In pairs: net cross effect = gross + wealth term

Class 6, slide 21, at its start: $p = (2, 3)$, $w = 12$. x_2 from the budget. $\sqrt{x_1} + \sqrt{x_2}$: the CES with $\rho = \frac{1}{2}$.

u	$x_1(p, w)$	gross $\frac{\partial x_1}{\partial p_2}$	wealth term $x_2 \frac{\partial x_1}{\partial w}$	net $\frac{\partial h_1}{\partial p_2}$
$\min\{x_1, x_2\}$	$\frac{w}{p_1 + p_2}$	-0.48	$+0.48$	0
$x_1 x_2$	$\frac{w}{2p_1}$	0	$+0.5$	0.5
$\sqrt{x_1} + \sqrt{x_2}$	$\frac{w\, p_2}{p_1(p_1 + p_2)}$	$+0.48$	$+0.48$	0.96

Four minutes.

Good 1 normal: net $\geq$ gross. The gap of class 6 is $x_2\, \partial x_1/\partial w$.

Slide 13 In pairs: net cross effect = gross + wealth term

Class 6's slide 21 said that gross and net cross effects can disagree and that the gap is a wealth effect; this table measures the gap for the same three consumers at the start point. The recipe is slide 9 with $\ell = 1$ and $k = 2$: the net effect is the gross effect plus $x_2\, \partial x_1/\partial w$, the quantity of the good whose price moves times the wealth slope of the good we watch.

Leontief: a dearer good 2 makes her poorer and she buys fewer pairs, a negative gross effect; compensated, she keeps buying the same pairs, so the net effect is zero, class 6's neither. For $x_1 x_2$ nothing happens gross, since half of w goes on good 1 whatever p_2, so the whole net effect is the wealth term; for the CES consumer the two parts add. In every row the wealth term is positive because good 1 is normal, so the net effect sits above the gross one, and with two goods it is never negative (class 6, slide 20). For a homothetic consumer the wealth term is $x_1 x_2/w$, which is why rows 1 and 3 give the same 0.48.

The slips: x_1 in the wealth term instead of x_2, or $\partial x_2/\partial w$ instead of $\partial x_1/\partial w$; and the sign of the gross effect for Leontief.

(optional) The Slutsky equation in elasticities

$\varepsilon_{11} = (\partial x_1/\partial p_1)\, p_1/x_1$, $\varepsilon_{1w} = (\partial x_1/\partial w)\, w/x_1$, $\varepsilon^h_{11} = (\partial h_1/\partial p_1)\, p_1/h_1$;
budget share $b_1 = p_1 x_1/w$.

(1) Show: $\varepsilon_{11} = \varepsilon^h_{11} - b_1\, \varepsilon_{1w}$.

Slutsky (slide 9), $\ell = k = 1$, times p_1/x_1:

$$\frac{\partial x_1}{\partial w}\, x_1\, \frac{p_1}{x_1} = \varepsilon_{1w}\, b_1$$

(2) $\varepsilon^h_{11} = -0.2$. For which ε_{1w} is good 1 Giffen if $b_1 = 0.4$? If $b_1 = 0.02$?

$\varepsilon_{1w} < -0.5$; $\varepsilon_{1w} < -10$:
Small budget share: Giffen needs an absurd ε_{1w}.

MWG (3.G.3), p. 71; elasticities as in MWG 2.E, p. 27; budget share b_ℓ, p. 28. Invented numbers in (2).

Break

Back at 11:00

Slide 14 (optional) The Slutsky equation in elasticities

Slide 9's own-price form made unit-free. In elasticities the wealth correction comes weighted by the budget share, and the reason is in words: a one per cent rise in p_1 makes her poorer by b_1 per cent of her wealth, since that is the share she spends on good 1, and the wealth elasticity turns that into a per cent change in x_1. On slide 12's rice the check reads $2 = 0 - \frac{1}{2} \cdot (-4)$.

That weight is what makes Giffen goods rare: the compensated elasticity ε^h_{11} is never positive, so a positive market elasticity needs the wealth term $-b_1\varepsilon_{1w}$ to be positive and to outweigh it, and with a small share the wealth elasticity would have to be absurd; a staple that takes half the budget, like the rice, can manage it. The same weighting is why the wealth effect on a 3-dollar mug is only cents on slide 22. The slips: dropping b_1, a plus sign on the wealth term, and reading inferior as Giffen: at $b_1 = 0.4$ a wealth elasticity of -0.3 is inferior and still not Giffen.

Slide 15 Break

Ten minutes.

The Slutsky matrix: class 6's $D_p h$, built from observable demand

$S(p, w)$: $s_{\ell k} = \dfrac{\partial x_\ell}{\partial p_k} + \dfrac{\partial x_\ell}{\partial w} x_k$

only Walrasian demand: observable in principle

$$S = \begin{pmatrix} -\frac{3}{2} + 3 \cdot \frac{1}{4} & 0 + 2 \cdot \frac{1}{4} \\ 0 + 3 \cdot \frac{1}{6} & -\frac{2}{3} + 2 \cdot \frac{1}{6} \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{3} \end{pmatrix}$$

$= D_p h(p, 6)$: class 6, slide 20

negative semidefinite symmetric $S\,p = 0$

Demand whose S is not symmetric or not negative semidefinite comes from no rational preferences.

MWG Prop. 3.G.2, pp. 69–70; 3.G, pp. 72–73: $S(p, w) = D_p h(p, u)$ (Prop. 3.G.3).

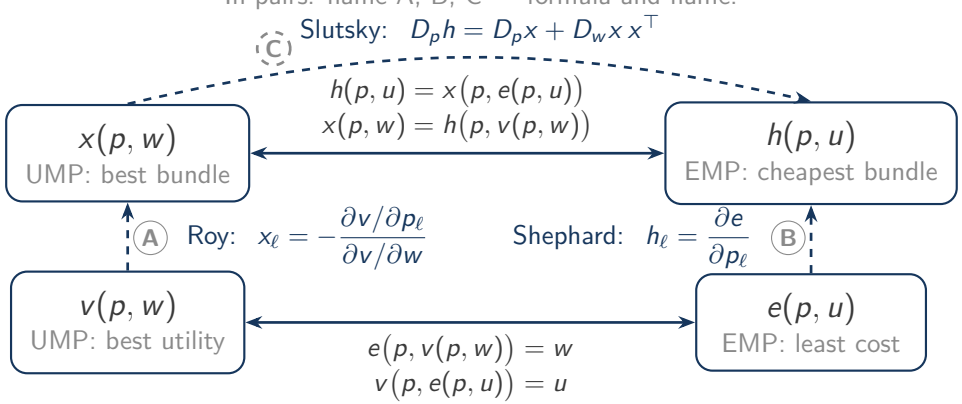
Slide 16 The Slutsky matrix: class 6's $D_p h$, built from observable demand

Slide 9 rearranged: every compensated slope can be computed from market demand alone, its price slopes, its wealth slopes and the bundle, all observable in principle, while h and e never are. Doing it for $x_1 x_2$ returns exactly the matrix of class 6's slide 20, which there came from the second derivatives of e: class 6's promise kept. Because $S = D_p h = D_p^2 e$, it has three properties, each for its own reason: own effects never positive, because e is concave in prices; equal cross effects, because the cross second derivatives of e are equal; and $S\,p = 0$, because h does not move when all prices scale.

Why it matters: these are restrictions on demand data that need no utility function to state. A demand whose S fails symmetry cannot come from any preferences (MWG p. 73); that nobody would guess the symmetry from economics alone is Samuelson's remark. Gross effects have no such property, as class 5's rice and meat showed.

In pairs: the map of x, v, h, e — name the three derivative arrows

In pairs: name A, B, C — formula and name.



Every arrow is from classes 4–7; the dashed ones are today's and last week's.

MWG Figure 3.G.3, p. 75 (a similar figure: Deaton and Muellbauer 1980); Props. 3.G.1, 3.G.3, 3.G.4.

Slide 17 In pairs: the map of x, v, h, e — name the three derivative arrows

MWG's Figure 3.G.3. The solid arrows are class 6's identities, which say that the two problems pick the same bundle and that e and v undo each other. The dashed arrows are the three derivative results of classes 5 to 7, and each has a reason for its shape: Roy's identity is a ratio of two slopes because v is in utils and the wealth slope converts them to euros; Shephard's lemma is one slope because e is already in euros; the Slutsky equation gives only the price slopes of h from x, never h itself, because h needs a utility level that x does not carry.

Why the map matters: from v, e or h all four functions can be reached by following arrows, while from x the arrows give only the slopes of h; that is the method of the problem sets. The slips: a minus sign or a ratio in Shephard's lemma, Roy's identity without its denominator, and reading arrow C as delivering h.

Money measures of welfare

MWG 3.I

slide 18 of 26

Slide 18 Money measures of welfare

Part 2: the money value of a price change, compensating and equivalent variation. MWG 3.I.

Price the change in euros: EV at the old prices, CV at the new

money metric: $e\big(\bar{p}, v(p, w)\big)$

utility in euros at fixed prices $\bar{p}$

equivalent variation: $\mathrm{EV} = e(p^0, u^1) - w$

at the old prices, the change in wealth as good as the price change: $v(p^0, w + \mathrm{EV}) = u^1$

compensating variation: $\mathrm{CV} = w - e(p^1, u^0)$

at the new prices, what can be taken after the change to restore u^0: $v(p^1, w - \mathrm{CV}) = u^0$

$u^0 = v(p^0, w), \quad u^1 = v(p^1, w)$

EV and CV: both negative when she loses

Both rank p^0 and p^1 correctly; their euros differ.

MWG 3.I, pp. 81–82: Figure 3.I.1; (3.I.1), (3.I.2), after Hicks (1939).

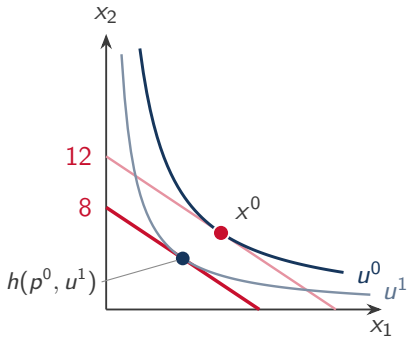
slide 19 of 26

Slide 19 Price the change in euros: EV at the old prices, CV at the new

Utility numbers mean nothing by themselves (class 3); euros do. The money metric fixes a price vector $\bar{p}$ and asks how much money she would need at $\bar{p}$ to be as well off as she is at (p, w). It is a utility function for the same preferences, increasing in v, and it does not change when u is relabelled, because e sees only the curve (class 6, slide 12). Two natural choices of $\bar{p}$ give Hicks's two measures: the old prices for EV, the new prices for CV.

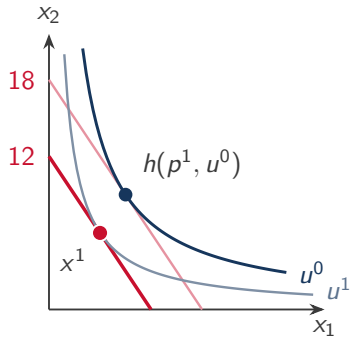
EV answers: what change in wealth, with no change in prices, would do to her exactly what the price change does. CV answers: after the change, how much could be taken from her to bring her back to her old utility, a negative number meaning she must be given money. Both carry the sign of the welfare change, so both rank the two price vectors correctly; their sizes differ because they price utility at different prices. The trap is to swap the prices and the utility levels inside e: EV uses the old prices with the new utility, CV the new prices with the old utility.

Good 1 to 4.5: EV $= -4$ euros, CV $= -6$ euros



EV: old prices

$e(p^0, u^1) - w = 8 - 12 = -4$



CV: new prices

$w - e(p^1, u^0) = 12 - 18 = -6$

Old prices value her loss at 4 euros, new prices at 6: EV $>$ CV, good 1 is normal.

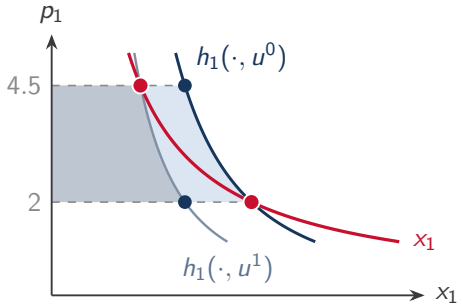
MWG Figure 3.I.2, p. 82; $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3)$, $p^1 = (4.5, 3)$.

Slide 20 Good 1 to 4.5: EV $= -4$ euros, CV $= -6$ euros

The same event, class 6's price rise, priced twice. Left, at the old prices: the utility she ends up with, u^1, could have been reached at p^0 for 8 euros, so the price rise did her as much harm as losing 4 of her 12. Right, at the new prices: getting back to u^0 costs 18 (class 6), 6 more than she has. Both numbers are negative in MWG's convention because she loses.

Why they differ: good 1 is normal, and valuing her loss at the old, cheap price of good 1 makes the loss look smaller, so EV $>$ CV; for an inferior good the order reverses (MWG p. 83). The fixed basket of class 6 said she needs 7.5 more; the truth, CV, is 6: concavity of e makes the fixed-basket number an upper bound on the loss. MWG's Figure 3.I.2 reads the euros on the vertical axis because it sets $p_2 = 1$; here $p_2 = 3$, so read the cost labels on the lines.

$-$CV $= 6$ and $-$EV $= 4$ are areas left of Hicksian demand



$-\text{CV} = \int_2^{4.5} h_1(p_1, 3, u^0)\, dp_1 = 6$

$-\text{EV} = \int_2^{4.5} h_1(p_1, 3, u^1)\, dp_1 = 4$

$\int_2^{4.5} x_1(p_1, 3, 12)\, dp_1 = 6 \ln 2.25 \approx 4.87$

market demand: in between

$-$CV: both shades $-$EV: dark

Shephard turns a change in e into an area under compensated demand.

MWG 3.I, pp. 83 and 89: (3.I.3), (3.I.4), (3.I.8); Figure 3.I.3.

Slide 21 $-$CV $= 6$ and $-$EV $= 4$ are areas left of Hicksian demand

Why areas: by Shephard's lemma $\partial e / \partial p_1 = h_1$, so the change in e as p_1 moves from 2 to 4.5 at a fixed utility is the integral of compensated demand between the two prices, the area to the left of that Hicksian curve. CV holds utility at u^0 and uses the outer curve; EV holds it at u^1 and uses the inner one. The market demand curve starts on the u^0 curve, where 12 euros buy u^0, and ends on the u^1 curve, where they buy u^1, so for a normal good its area lies between the two: MWG's area variation, which overstates one measure and understates the other.

When do all three agree? If good 1 has no wealth effect, for instance preferences quasilinear in the other good, the three curves coincide and CV $=$ EV $=$ the change in Marshall's consumer surplus. The trap is to take the market-demand area as the exact welfare change; it is exact only without a wealth effect, and slide 24 shows what the wrong curve would do for a tax.

Buyers pay a CV, choosers take an EV; 7.12 is no wealth effect

group	in our theory	median
Buyers	CV of getting a mug: pay it and stay as well off	\$2.87
Choosers	EV of getting a mug: cash as good as the mug	\$3.12
Sellers	the choosers' choice: mug or cash	\$7.12

EV − CV: a wealth effect; for a 3-dollar mug, cents

Buyers vs choosers: as theory says. Sellers vs choosers: no wealth effect can do that; owning the mug changed its value.

Kahneman, Knetsch & Thaler (1990), JPE 98(6), p. 1339: "these results eliminate any form of income effect as an explanation".

In pairs: good 1 falls to 0.5: what is the gain worth?

$u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (0.5, 3)$.

question	answer
new bundle x^1 and utility u^1	$(12,\ 2)$, $u^1 = 24$
$h(p^1, u^0)$: split x_1 into substitution and wealth	$(6,\ 1)$; x_1: $3 \to 6 \to 12$ substitution $+3$, wealth $+6$
$\text{CV} = w - e(p^1, u^0)$	$12 - 6 = 6$
$\text{EV} = e(p^0, u^1) - w$	$24 - 12 = 12$

Five minutes.

A gain: both positive, and EV > CV again: good 1 is normal.

Slide 22 Buyers pay a CV, choosers take an EV; 7.12 is no wealth effect

The change is getting a mug. A buyer, without a mug, pays at most the sum that leaves her as well off as before: that is the CV of getting the mug, what can be taken after the change to restore her old utility. A chooser names the cash that is exactly as good as the mug: the EV of the same change. A seller faces the chooser's choice, mug or cash, so theory makes sellers and choosers equal, and buyers a little lower, since EV $\geq$ CV for a normal good by a wealth effect.

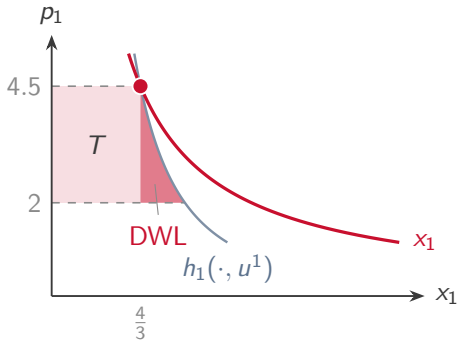
The data fit the second prediction and reject the first. Buyers' \$2.87 lies below choosers' \$3.12, as theory predicts, and the 25-cent gap is small, as it should be: EV − CV is what being about three dollars richer does to a mug's money value, cents rather than dollars; and the medians come from different students, so part of a small gap is chance. Sellers' \$7.12 is more than twice the choosers', yet their positions were identical, so theory allows no gap; even counting the mug as wealth, being a few dollars richer would have to raise a mug's value by four dollars. The authors call it the endowment effect: owning the object changed its value. That lies outside this course's consumer theory, where preferences do not depend on what you happen to own.

Slide 23 In pairs: good 1 falls to 0.5: what is the gain worth?

The same consumer, now with good 1 getting cheaper, so every number should come out as a gain. Line 1 is class 4's demand. Line 2 is the cheapest bundle on the old curve at the new prices, which class 6's envelope slide already found at $p_1 = 0.5$; the substitution step runs towards the cheaper good and the wealth step adds more of it, because good 1 is normal. Line 3 prices the old utility at the new prices: she now needs far less than 12, and the surplus is what a planner could take. Line 4 prices the new utility at the old prices: the wealth she would have needed to be this well off without the price fall.

Both measures are positive, and EV > CV again, for the same reason as on slide 20: the good whose price fell is normal. As areas they are the regions left of the two Hicksian curves between 0.5 and 2, with the market area between them. The slips: the wrong utility level inside e, EV computed at the new prices, and taking the new market bundle for the compensated one.

Tax good 1 by 2.5: it raises 3.33, costs her 4: deadweight loss 0.67



tax $t = 2.5$ per unit: p_1: $2 \to 4.5$

revenue $T = t\,x_1(p^1, w) = 2.5 \cdot \frac{4}{3} \approx 3.33$

her loss $-\text{EV} = 4$

deadweight loss $= 4 - 3.33 \approx 0.67$

The tax takes more from her than it brings in: a triangle left of Hicksian demand.

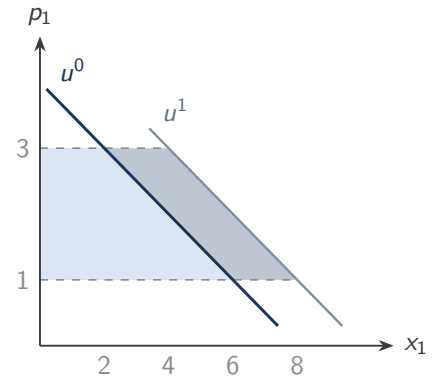
MWG Example 3.I.1, pp. 84–85, (3.I.5), Figure 3.I.4(a); own numbers.

Slide 24 Tax good 1 by 2.5: it raises 3.33, costs her 4: deadweight loss 0.67

Read the price rise as a per-unit tax of 2.5 on good 1. The state collects the tax on what she buys after the change. Her loss, measured as EV, is larger than that revenue: if the state took the same sum as a lump sum instead, prices would stay at p^0 and she would keep more than the 8 euros that her post-tax utility costs at p^0, so she would be better off and the state no poorer. The gap is the deadweight loss of the tax, MWG's Example 3.I.1.

In the figure the EV area of slide 21 splits into the revenue rectangle and the sliver beside it. The sliver is never negative, because compensated demand never rises in its own price, and it is zero only when she cannot substitute, as for Leontief, where it has no width. MWG measures it against the Hicksian curve at u^1, not the market curve; the market curve would give the wrong triangle, which is the trap.

(optional) CV and EV from two compensated demand curves



p_1: $1 \to 3$; p_2, w fixed.

$h_1(p_1, u^0) = 8 - 2p_1$, $h_1(p_1, u^1) = 10 - 2p_1$

(1) CV and EV?

CV $= -8$ (light), EV $= -12$ (both shades)

(2) Normal or inferior?
Is EV $>$ CV, as on slide 20?

Inferior: more of it at the lower u^1.
No: EV $<$ CV.

MWG 3.I, p. 83: (3.I.3), (3.I.4), Figure 3.I.3. Own numbers.

Slide 25 (optional) CV and EV from two compensated demand curves

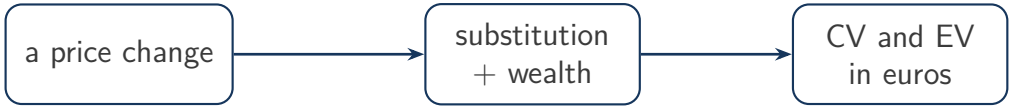
No utility function this time, only the two compensated demand curves, and that is all slide 21's recipe needs: each measure is the area to the left of one line between the two prices, here a trapezoid, CV on the line at the old utility, EV on the line at the new one, and both negative because she loses.

Which one is larger depends only on where the u^1 line lies. The price rise lowers her utility, and here the lower-utility line lies to the right: poorer, she buys more good 1 at every price, which is what inferior means ($\partial h_1/\partial u$ has the sign of $\partial x_1/\partial w$), so the EV region contains the CV region, the reverse of slide 21's shading and of slide 20's order, which held for a normal good only. Her market demand still falls, from 6 at $p_1 = 1$ to 4 at $p_1 = 3$, less than along either line: inferior but not Giffen, slide 10's picture reversed. The slips: EV $>$ CV taken as a rule, EV read off the u^0 line, and positive signs for a loss.

Today in one line

Slutsky:
$$\partial h/\partial p = \partial x/\partial p + (\partial x/\partial w)\, x^\top$$

e at p^1 or p^0:
areas left of h



Wednesday: Uncertainty I: lotteries, the independence axiom, the expected utility theorem and the uniqueness of the vNM utility function.
Money lotteries, risk aversion, certainty equivalent, risk premium.
Problem set 2: due at the start of class, Wednesday 21 October — on paper, one copy per group.

Read MWG 6.B–6.C (or Muñoz-Garcia ch. 5) for Wednesday 21 October.

Slide 26 Today in one line

In one line: a price change moves her along her curve, the substitution effect, never towards the dearer good, and shifts her budget, the wealth effect, whose sign depends on the good; that is the Slutsky equation, and it is why a Giffen good must be inferior. The matrix of substitution effects can be computed from market demand and must be symmetric and negative semidefinite. The map ties x, v, h and e together. In euros a price change is worth CV at the new prices or EV at the old, areas to the left of Hicksian demand, equal when there is no wealth effect.

Wednesday 21 October: uncertainty, lotteries, the independence axiom, the expected utility theorem and the uniqueness of the vNM utility function; money lotteries, risk aversion, certainty equivalent, risk premium. Reading: MWG 6.B–6.C, or Muñoz-Garcia chapter 5.

Problem set 2 is due at the start of class on Wednesday 21 October, on paper, one copy per group.